Embodied Field Intelligence: Real-Time Optimal Control from Local Belief and Value Fields

Joshua Winters
Independent Researcher

Abstract

We study an embodied agent constrained to its founding premise: it perceives only a 5×5 window, carries only internal state, and must act competently in real time with *zero training episodes*. We show that this regime admits a principled architecture built from exactly two local field computations: a *belief field* — a log-odds Bayes filter over target locations whose prediction step is masked diffusion and whose correction step includes negative evidence — and a *value field* — the desirability recursion of a linearly-solvable MDP (LMDP), computed in the log domain as a radius-1 stencil operation, with repulsors entering as state costs and the optimal policy given in closed form as a softmax over neighboring values. Against the classical recipe for such agents — superpose attractor and repulsor potentials and follow the gradient — which we implement on the identical substrate as a baseline, this formulation repairs structural failures we demonstrate head-to-head: value attenuates *linearly* rather than exponentially with distance (success 98.9% at both 15×15 and 30×30, versus the baseline's 33% falling to 27%), aversive stimuli become path costs the planner routes around (rather than subtracted potentials an attractor can outbid), and exploration becomes an expected-information-gain reward computed from belief entropy. A single risk parameter λ, set continuously by an affect system, subsumes the three ad-hoc mechanisms potential-field designs otherwise require (action temperature, a max-plus "semiring flip," and noise scaling), and infinite state costs yield *exact* barrier guarantees. On a stochastic foraging benchmark the agent attains 0.97 of a full-observability A^* oracle's normalized score with zero training, against 0.50 for a memoryless greedy agent and -0.02 for tabular Q-learning after $2{,}000$ training episodes. Because cognition is an amortized fixed-point computation, thinking is measurable: performance collapses at zero sweeps per tick and the agent's reaction latency to a revealed stimulus at distance d fits $0.99\,d/\kappa$ ticks ($R^2 = 0.999$) — an internal "speed of thought" of one cell per sweep, which doubles as a falsifiable locality certificate. In pre-registered non-stationarity experiments the agent re-values a swapped reward in 5 steps (baselines: 69–83) and is the only contender with positive return under continual regrowth. An egocentric variant with no global position or world-size access matches the world-frame agent via dead reckoning plus window-overlap visual odometry ($13\times$ lower pose drift than dead reckoning alone). Code, tests, and per-figure reproduction commands are included.

1 Introduction

Most competent artificial agents are trained: a policy is optimized offline against an environment distribution and then deployed. This paper develops the opposite regime and takes it seriously as a design constraint. Our agent is a closed box with a narrow sensory window and an internal state; everything it knows it must have seen, everything it does must be computed *during its lifetime* by bounded per-tick internal dynamics, and nothing may be learned by gradient descent across episodes. The interest of the regime is twofold. Practically, it is the regime of online adaptation: an agent that never assumed stationarity has nothing to lose when the world changes. Scientifically, it forces a clean answer to the question *what internal quantities are worth computing when computation is local and time is real*.

The classical answer for such an agent is a collection of hand-designed spatial fields: diffused "scent" toward observed targets, a repulsive visit trail, novelty and frontier attractors, and hazard repulsors, superposed linearly into a potential whose local gradient selects actions (Khatib, 1986; Brooks, 1991; Dorigo et al., 1996). That recipe is interpretable and cheap — we instantiate it in full, on the identical substrate, as our principal baseline — but it carries three structural defects that our experiments expose head-to-head (Section 6.3). Diffused gradients attenuate exponentially, so competence collapses with grid size; repulsors, being *subtracted potentials*, can always be outbid by a strong attractor, so safety claims are probabilistic at best; and taming the resulting dynamics demands a sprawl of interacting patches — temperature schedules, mode switches, anti-oscillation penalties, gadget exploration fields — none of which has a defined objective. *Embodied Field Intelligence* (EFI), the architecture developed here, keeps the substrate — everything is still a scalar field updated by radius-1 stencils — and replaces its physics.

The replacement rests on one observation: the two things such an agent needs, an estimate of *what is out there* and a ranking of *what to do about it*, are both fixed points of local operators, and both have exact, classical formulations.

1. **Beliefs are a Bayes filter that was already almost there.** Diffusing a field seeded at observed targets is an informal posterior. Making it a log-odds occupancy filter (Elfes, 1989; Thrun et al., 2005) costs one line — and gains *negative evidence*: observing an empty cell now erases stale belief, which the scent heuristic could never do.
2. **Value iteration is literally masked diffusion.** In Todorov's linearly-solvable MDPs (Todorov, 2006, 2009) the desirability $z = e^{V/\lambda}$ satisfies a *linear, local* fixed-point equation: $z(v) = e^{-q(v)/\lambda} \cdot$ (neighbor average of z). The scent-diffusion loop EFI always ran was an unwitting approximation of optimal control. Running the recursion exactly, in the log domain, converts the architecture's central heuristic into a theorem-backed computation — and the log-domain readout makes value *linear* in distance, which is what removes the scaling ceiling.

Everything else in the architecture then falls into place as corollaries. Repulsors (visit trail, hazards, protective membranes, pain) enter the recursion as state costs $q(v)$, which is the mathematically correct way for penalties to shape paths; a cell with $q = \infty$ is an *exact* barrier no attractor can outbid. The LMDP's temperature λ is simultaneously the planning risk parameter (its $\lambda \to 0$ limit is the max-plus/worst-case semiring) and the closed-form optimal policy's softmax temperature, so the affect system controls one dial instead of three mechanisms. Exploration stops being a hand-tuned gadget: with calibrated beliefs, per-cell entropy is free, and expected information gain pooled over the sensory window is a reward like any other, injected into the same recursion. And because the value field is a warm-started fixed-point computation, *thinking has units* — sweeps per tick — with measurable consequences we exploit throughout the experiments.

Contributions.

1. **A belief–value reformulation of field-based control** (Section 4): log-odds Bayes-filter beliefs with negative evidence, and LMDP value iteration computed as a local log-domain field operation with repulsors as state costs and the closed-form softmax policy.
2. **One-dial safety** (Section 5): affect sets λ; pain-monotone action choice and an *exact* barrier certificate ($q{=}\infty \Rightarrow$ entry probability 0 for every $\lambda > 0$), strengthening the probabilistic guarantees of the potential formulation. Empirically: zero violations in 500-step adversarial walks across three orders of λ.
3. **Cognition with units** (Sections 5 and 6): real-time control as tracking of a moving fixed point, with a contraction bound validated on-trajectory (100% of quiescent windows within the bound), thinking-rate curves ($\kappa{=}0$ collapses to 5–8% success; $\kappa{\geq}1$ saturates), and a measured internal speed of thought of 1.01 cells per sweep ($R^2 = 0.999$) that doubles as a locality regression test.
4. **Empirics with floors and ceilings** (Section 6): 0.97 normalized score against an A^* oracle at zero

training; scale-invariant success; ablations showing the load-bearing component is the value recursion itself, not any single field; pre-registered non-stationarity results (revaluation in 5 steps vs. 69–83; the only positive return under regrowth), including an adverse finding we report as prominently as the favorable ones.

5. **Egocentric embodiment** (Section 6.8): a variant with no access to global position, world size, or ground-truth maps, driven by an efference copy plus window-overlap visual odometry; it matches the world-frame agent and cuts odometry drift $13\times$ under 10% actuation slip.

2 Related Work

Potential fields and stigmergy. Artificial potential fields for navigation date to Khatib (1986); their known pathologies (local minima, attractor/repulsor arms races) stem from composing *forces* rather than solving a control problem. Pheromone-like trail memory is classical stigmergy (Dorigo et al., 1996). EFI retains both ingredients but moves them inside a value recursion: trails become transit costs, and the composed field is an optimal value function rather than a superposition heuristic.

Linearly-solvable optimal control. Our control law is Todorov's LMDP desirability iteration (Todorov, 2006, 2009), closely related to path-integral control (Kappen, 2005) and to entropy-regularized RL. Our contribution is architectural: identifying the desirability recursion as a *drop-in replacement for diffusion* in field-based agents, running it in the log domain on the agent's own believed map, warm-started across ticks, with beliefs supplying the reward injection.

Occupancy grids and probabilistic robotics. The belief fields are log-odds occupancy filters (Elfes, 1989; Thrun et al., 2005); the egocentric variant's pose maintenance (dead reckoning, window-overlap egomotion estimation, anchor matching) is a deliberately minimal, fully local instance of the classical localization stack.

Safe control. Control barrier functions (Ames et al., 2017) certify invariance of safe sets. Our discrete analogue is blunt but exact: an infinite state cost makes the closed-form policy's probability of entering the forbidden set identically zero whenever an alternative exists (Proposition 2), a property subtracted-potential formulations cannot have.

Active inference. The free-energy framing (Friston, 2010) also couples belief updating, expected information gain, and affect-like precision parameters. EFI can be read as a tractable, non-variational instance: every quantity is a closed-form field update, the epistemic bonus is window-pooled binary entropy, and the precision analogue λ has an exact optimal-control semantics. We view the correspondence as convergent structure rather than derivation.

Learning-based agents. Against deep and tabular RL (Sutton and Barto, 2018) our claim is not asymptotic performance but the zero-training operating point and non-stationary robustness (Section 6.7). Neural cellular automata (Mordvintsev et al., 2020) learn the field dynamics themselves; our dynamics stay hand-specified and interpretable, and the only learned components (valences, a count-based local world model) are online and gradient-free, in the spirit of Brooks (1991).

3 Setting

FORAGEWORLD is an $H\times W$ gridworld with i.i.d. walls (density 0.12), n_A appetitive targets ($r_A{=}{+}1$), n_B aversive targets ($r_B{=}{-}0.8$), step cost -0.01 and bump penalty -0.02. The agent observes a 5×5, 4-channel binary window (walls, A, B, self); out-of-bounds cells read as walls (impassability is observable, so world extent is discoverable rather than given). Actions are the 4-neighborhood moves. Episodes end at a step cap or when all targets are collected. Non-stationary variants (Section 6.7) add target regrowth, periodic drift, and a mid-episode reward swap. The task is a POMDP; the agent never accesses the world map, the target list, or (in the egocentric variant, Section 6.8) its own coordinates.

4 Method: two fields and a dial

The agent's internal state consists of per-channel belief fields L_t^c, a visit trail, a value field V_t, a handful of affect scalars, and (optionally) a learned table of local transition rules. One tick runs: *sense* (belief correction), *predict* (belief diffusion), *think* (κ value sweeps), *act* (neighbor softmax), *integrate* (efference copy, valence update on pickups). Algorithm 1 summarizes; each stage is a radius-1 stencil or a per-cell operation.

4.1 Belief fields: a Bayes filter in log-odds

For each target channel $c \in \{A, B\}$ the agent maintains $L^c(v) = \log \frac{P(\text{target at } v)}{P(\text{no target at } v)}$, initialized at a prior $\ell_0 = -4$ ($p \approx 0.018$). The **correction** step adds ℓ_+ at observed targets and $\ell_- < 0$ at cells *observed empty*, clamped to $[-8, 8]$; increments (± 12) deliberately exceed the clamp so a single noiseless observation saturates belief in either direction, while the clamp keeps any belief revisable. Negative evidence is the operative difference from scent fields: a disproven peak dies in one glance instead of decaying over hundreds of ticks. The **prediction** step diffuses the probability map $p = \sigma(L)$ through the agent's *known* walls (masked diffusion; probability mass cannot leak through mapped structure) and relaxes L toward the prior — uncertainty grows only where the agent is not looking, at a rate gated by the learned world model (Section 4.5). Composition consumes probabilities weighted by *learned valences* w^c (updated online from experienced pickup rewards), not by environment constants the agent has no right to know.

4.2 Value as a local fixed point

Let $q(v) \geq 0$ collect state costs (per-step effort q_{step}, trail transit cost, hazard shells, membrane, pain, and $|w^c|\, p^c(v)$ for channels with *negative* learned valence) and let $R(v)$ collect expected arrival rewards ($w^c p^c(v)$ for positive-valence channels, plus the epistemic bonus of Section 4.4; the belief prior supplies a small optimism floor everywhere — unexplored cells might pay at prior rate). For an LMDP with uniform passive dynamics over passable neighbors $N(v)$, the desirability $z = e^{V/\lambda}$ satisfies a linear local fixed-point equation; in the log domain one sweep is

$$V \;\leftarrow\; \max\Big(\underbrace{\lambda \log \tfrac{1}{|N(v)|} \sum_{u \in N(v)} e^{V(u)/\lambda}}_{\text{lse}_\lambda\text{: neighbor average of } z},\; R \Big) \;-\; q, \qquad V\big|_{\text{walls}} = -\infty, \tag{1}$$

implemented with the same 4-shift stencil as the diffusion operator, numerically stable by max-subtraction, with a finite sentinel for $-\infty$. The optimal policy is closed-form (Todorov, 2006): $\pi^*(v \to u) \propto z(u) = e^{V(u)/\lambda}$, sampled exactly by Gumbel-max over neighbor values — there is no separate action-selection machinery, no momentum, and no anti-oscillation hack; correct values remove the pathologies those hacks patched.

Three consequences of Equation (1) do real work. *(i) Order of operations.* Rewards are injected *before* costs are charged: arriving somewhere to collect its reward still pays that cell's cost. The reversed order ($\max$ after $-q$) lets a cell's reward erase its own cost; empirically this single transposition was worth a $14\times$ reduction in aversive-target contact and moved mean return from -1.67 to -0.12 (Section 6.3). *(ii) Costs, not negative attractors.* A subtracted repulsor double-counts geometry and loses any bidding war against a sufficiently strong attractor; a state cost is charged along the planner's own path and, at $q = \infty$, cannot be outbid at any price (Proposition 2). Cost weights are accordingly small ($q_{\text{trail}} = 0.08$, an order below the

0.6 the potential baseline needs for the same repulsion: a penalty that strong, charged as a path cost, would price a corridor above the total available reward and trap the agent behind its own trail). *(iii) Warm starts.* V_t persists across ticks; each tick runs κ sweeps (default 3; $H+W$ once at episode start — the agent orients before moving). Real-time viability rests on the contraction property analyzed in Section 5.

4.3 One dial: affect sets λ

λ appears twice in Equation (1)'s ecosystem — as the planning risk parameter and as the policy temperature — and they are the *same* λ. Its limits are the two modes a potential-field design must hard-code separately: $\lambda \to 0$ turns lse_λ into $\max$ (the tropical/max-plus semiring: worst-case, shortest-path planning) and the policy greedy; large λ is diffusive and exploratory. The affect system therefore needs one output:

$$\lambda_t = \mathrm{clip}\big(\lambda_0\,(1 + k_a\,\mathrm{arousal}_t - k_p\,\mathrm{pain}_t),\ \lambda_{\min}, \lambda_{\max}\big). \tag{2}$$

Where the potential baseline needs a temperature schedule, a discrete linear→max-plus flip at a pain threshold, and a noise cap — three interacting mechanisms with a behavioral cliff at the flip — the value formulation needs none, and behavior is continuous in pain (verified by test). Because the LMDP charges a KL control cost of $\lambda \log |N|$ per step, λ must be small relative to reward scale; the *reach budget* of a reward r is $d^* \approx r/(q_{\text{step}} + \lambda \log 4)$, an explicit dial (defaults: λ_0=0.02, $d^* \approx 26$ cells per unit reward) where diffused potentials have an implicit exponential ceiling.

4.4 Exploration as expected information gain

Calibrated beliefs make uncertainty a first-class, local quantity: $u(v) = H_2(p^A(v)) + H_2(p^B(v)) + w_{\text{map}}\,[v \text{ unseen}]$ (binary entropies plus one unresolved occupancy bit; walls carry none). The epistemic reward is its mean over the sensory window centered at v — "what I would learn standing there" — scaled by an affect-modulated weight, $\beta_t = \beta_0(1 + k_c\,\mathrm{arousal})(1 - k_f\,\mathrm{pain})$, and injected into Equation (1) like any other reward. One principled term does the job for which potential-field designs carry two hand-tuned exploration gadgets (novelty and frontier fields, both present in our baseline); exploration–exploitation is the single scalar β, and a frightened agent verifiably stops sightseeing. Self-predicted novelty is deliberately *excluded* from the reward channel: prediction error deposited at the agent's own trailing cells, if injected as an absorbing reward, makes the agent plan back to where it just was — a self-trap that in early experiments drove 30×30 success to 0%.

4.5 A world model that earns its keep

The only learned structure beyond scalar valences is a count-based table of local transition rules, mapping each ($3\times3\times4$-channel neighborhood, action) pair to counts over the cell's next state, updated from the overlap of consecutive windows ($\sim$9 cells per tick), with every 10th transition held out for scoring. We chose it over the Hebbian prototype learners often proposed for such substrates (Oja/BCM-style (Oja, 1982)) because prediction gives the learner a *defined, measurable objective*; it earns its keep through three jobs. Its prediction error *is* the surprise signal driving arousal (unfamiliar contexts score maximal surprise, so novelty is arousing before any rule exists); its *static-confidence* (fraction of confident predictions where the world, excluding the agent's own channel, did not change) gates the belief prediction step, so memory stops blurring exactly when the world is learned to be static; and rolled forward with sensors detached it is a primitive imagination. In a static world it reaches 100% held-out accuracy and static-confidence 1.0 within two episodes (995 rules), learns uniform structure in one shot, and is return-neutral (Table 2) — a bar an objective-free learner has no reason to clear.

Algorithm 1 One EFI tick (all operations radius-1 stencils or per-cell).

sense: correct L^c from the window (ℓ_+ at targets, ℓ_- at observed-empty cells); map walls
predict: diffuse $p^c{=}\sigma(L^c)$ through known walls; relax toward prior (both gated by static-confidence)
model: update transition counts from consecutive-window overlap; emit surprise
affect: update pain/arousal (EWMA of nociception, surprise, reward); set λ_t, β_t, membrane radius
compose: $q \leftarrow q_{\text{step}} + \text{trail} + \text{hazards} + \text{membrane} + \sum_{w^c<0} |w^c| p^c; \quad R \leftarrow \sum_{w^c \geq 0} w^c p^c + \beta_t\, \overline{u}$
think: κ sweeps of Equation (1), warm-started; forbidden cells ($q{=}\infty$) held at $-\infty$
act: sample $a \propto e^{V(u)/\lambda_t}$ over open neighbors; step environment
integrate: efference copy $\rightarrow$ pose (Section 6.8); on pickup: valence update, belief collapse at the cell

5 Theory

Proofs are elementary; their value is that they attach exactly to the implemented operators (each is enforced by a unit test).

Proposition 1 (Linear attenuation and reach)**.** *Along a corridor of degree n with uniform cost c, the fixed point of Equation (1) satisfies $V(d) = r - \alpha d$ with $c \leq \alpha \leq c + \lambda \log n$ up to transverse-mode corrections; in particular $\|\nabla V\|$ is independent of d within the reach budget $d^* = r/\alpha$.*

In z-space attenuation is exponential ($z(d) \propto e^{-\alpha d/\lambda}$, which underflows and whose gradient vanishes at range — the scent failure); the log-domain readout is what makes distant rewards steer. Measured: the fitted corridor slope lies in the predicted band, residuals from linearity < 0.01, and the near/far gradient ratio is $1{\pm}0.5$ over 25 cells.

Proposition 2 (Exact barrier)**.** *Let F be a forbidden set with $q|_F = \infty$ (implemented $V|_F = -$VBIG, excluded from propagation). For every $\lambda > 0$ and every state with at least one open neighbor outside F, the closed-form policy satisfies $\Pr[\text{enter } F] = 0$.*

Proof. $\pi^*(v \to u) \propto e^{V(u)/\lambda}$ and Gumbel perturbations are a.s. finite, so a $-\infty$ score loses to any finite alternative with probability 1. No finite attractor weight can outbid an infinite path cost, unlike a subtracted potential. □

If *all* open neighbors are forbidden the softmax degrades to least-bad and a per-tick deadlock counter records the event, making design violations observable rather than silent. Empirically: zero entries into a membrane ring over 500-step adversarial walks for λ spanning 0.005 to 0.1.

Proposition 3 (Pain-monotone choice)**.** *Write $V = V_0 - w\, q_{pain}$ with $w > 0$. As pain drives $\lambda \to \lambda_{\min}$, the chosen neighbor satisfies $q_{pain}(u_{chosen}) \leq \min_u q_{pain}(u) + \varepsilon(\lambda_{\min})$ with $\varepsilon(\lambda) = \lambda\,(\Delta G + \Delta V_0)/w \to 0$.*

A potential-field design approximates this with several interacting mechanisms (nonnegative repulsor weights, backtrack penalties, capped noise); here it is one limit of one formula, and action-distribution entropy decreases monotonically in pain by construction (verified over the full pain range).

Proposition 4 (Fixed-point tracking)**.** *The sweep operator is a contraction on the passable region with rate $\gamma < 1$ (any positive q and a reward floor suffice; an enclosed rewardless pocket has fixed point $-\infty$ and is excluded, which the belief-prior optimism floor guarantees in the controller). If the beliefs move the fixed point by at most ε per tick, κ warm-started sweeps track it within $\varepsilon\gamma^\kappa/(1-\gamma^\kappa)$ at steady state.*

Real-time cognition, in this architecture, is amortized tracking of a slowly-moving fixed point, licensed by the temporal coherence of the world (at most one window of evidence arrives per tick). The bound's scope

Table 1: Baselines: 200 episodes/agent (40 episodes $\times$ 5 seeds), identical seeds. EFI reaches 0.97 of the full-observability ceiling *with zero training*; tabular Q cannot generalize across worlds from raw windows at this budget — the point of the sample-efficiency comparison.

Agent	Mean return	Success	Normalized	Training episodes
Random walk	-2.795 ± 0.909	6.0%	0.00	0
Greedy-visible	-1.386 ± 0.797	22.5%	0.50	0
Tabular Q	-2.841 ± 1.546	0.5%	-0.02	2000
EFI	$\mathbf{-0.087 \pm 0.286}$	**96.5%**	**0.97**	**0**
A^* oracle (ceiling)	-0.000 ± 0.000	100.0%	1.00	0

matters: pickups and discoveries *jump* the fixed point discontinuously, and no steady-state bound covers a jump — the field re-converges at rate γ afterward, which is precisely the reaction-latency phenomenon measured in Section 6.6. On-trajectory measurements (Section 6.5) validate both the bound and its stated scope.

Remark 1 (The light cone)**.** *Every operator in Algorithm 1 is a radius-1 stencil iterated a declared number of times, so information propagates through the internal state at most κ cells per tick. An agent of this class cannot react to a stimulus at distance d in fewer than $\sim d/\kappa$ ticks without prior belief. This is a falsifiable architectural signature — any cached global shortcut would flatten the latency–distance line — and we measure it directly in Section 6.6.*

6 Experiments

All experiments use the public implementation; every table and figure lists its reproduction command in Appendix A, and each result is also enforced as a pytest gate. Unless stated: 17×17 grids, $n_A=2$, $n_B=4$, 200-step episodes, identical environment seed lists across contenders, agents persist (and keep adapting) across an evaluation's episodes. The EFI configuration is identical across all experiments — no per-task tuning.

6.1 Floors and ceilings

Table 1 evaluates against a random-walk floor, a memoryless greedy-visible agent ("do you even need fields"), tabular Q-learning trained for 2,000 episodes on the same distribution then frozen, and a full-observability A^* oracle ceiling (nearest-target tour, replanned each step). Normalized score is $(X - \text{random})/(A^* - \text{random})$.

6.2 Scale invariance

The potential-composition baseline — identical fields, beliefs, and substrate; linear superposition and gradient-following in place of the value recursion — scores 33.3% at 15×15, falling to 26.7% at 30×30: most of its competence is already gone at small scale, and what remains decays with size as exponential attenuation predicts. The value-recursion controller scores **98.9%** at *both* sizes (30 episodes $\times$ 3 seeds, step cap $0.9\,HW$), as Proposition 1 predicts — there is no range mechanism left to fail. The remaining size dependence is step cost, not competence.

Table 2: Ablations (15×15, 25 episodes × 3 seeds each, shared seeds). Differences among the top rows are within noise; the bottom block — the same fields under potential composition instead of the value recursion — is the story.

Configuration	Return	Success	B contacts	Coverage
Full system	-0.088 ± 0.256	98.7%	0.09	67.8%
Epistemic: frontier gadget	-0.117 ± 0.283	100.0%	0.15	68.1%
Epistemic: none	-0.243 ± 0.456	86.7%	0.12	65.3%
No visit trail	-0.125 ± 0.345	97.3%	0.11	46.1%
Scent instead of beliefs	-0.133 ± 0.381	97.3%	0.13	65.7%
Predictive schema off	-0.117 ± 0.312	100.0%	0.15	66.8%
Potential composition (baseline)	-1.643 ± 0.889	24.0%	0.72	40.7%
Potential composition, no trail	-1.600 ± 0.880	14.7%	0.33	23.4%

6.3 Ablations: the algebra is the load-bearing component

Table 2 ablates the system and its potential-composition baseline on shared seeds (15×15, 75 episodes each). Two contrasts are worth flagging. First, *which component is load-bearing depends on the algebra*: under potential composition the visit trail is the critical exploration mechanism (removing it drops success from 24.0% to 14.7% and coverage from 40.7% to 23.4%), whereas under the value recursion it is nearly dispensable for returns (98.7% → 97.3% success without it), correct values subsuming most of its anti-dithering role. Second, no single field is critical to the full system, but swapping the value recursion for potential composition — keeping every field identical — collapses success from 98.7% to 24.0%: **the algebra, not any field, is the load-bearing element**. The epistemic term is the largest single-field contributor (86.7% → 98.7% success, and it is what sustains adaptation in Section 6.7); beliefs-vs-scent and the predictive schema are return-neutral here (their value shows under non-stationarity and in the surprise channel, respectively) — we report this rather than claim otherwise. Separately measured: the reward-injection order of Equation (1) is worth 2.07 → 0.15 aversive contacts per episode, and belief fields beat scent by +0.24 return even under the *potential* composition (150 episodes) — negative evidence helps independently of the control law.

6.4 Thinking has units: the κ curves

κ (value sweeps per tick) is the thinking rate. Figure 1 shows its behavioral curve: κ=0 (act on the stale field) collapses to 5–8% success; κ=1 already recovers the ceiling. Two honest readings: thinking is load-bearing, and *this* task saturates at one sweep per tick — warm-started tracking is that effective, so resolving higher κ requires harder worlds. The instrument still sees what behavior cannot: the mean per-tick fixed-point residual falls smoothly with κ (0.157 → 0.017 from κ=1 to 8 at 15×15), exactly as contraction predicts.

6.5 The tracking bound, on trajectory

On an instrumented episode (15×15, κ=3; the true fixed point recovered by 200 extra sweeps on frozen inputs every 10 ticks): contraction rate $\hat{\gamma} = 0.750$ (median per-sweep residual ratio); fixed-point drift $\varepsilon = 0.082$ per window; Proposition 4 bound = 0.060; median measured tracking error **0.0039** — 15× inside the bound — with **100% of quiescent windows within the bound** (11/19). The other 8 windows are belief jumps (pickups, discoveries; drift > 3× median, transient median error 0.13), where a steady-state bound does not apply by its own hypotheses; we report both counts rather than claim the bound where it has no scope.

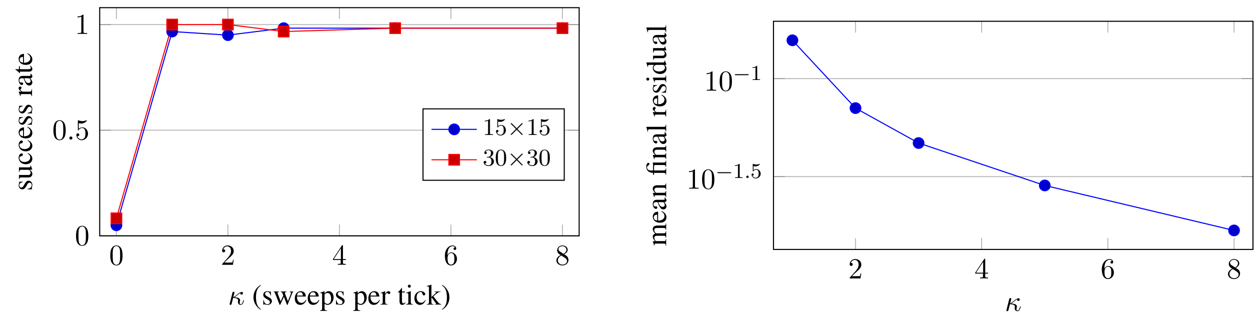

Figure 1: Thinking rate. *Left:* success vs. κ (20 episodes $\times$ 3 seeds per point): κ=0 collapses, $\kappa\geq1$ saturates. *Right:* per-tick fixed-point residual falls geometrically with κ (15×15) — the contraction is visible even where behavior has saturated.

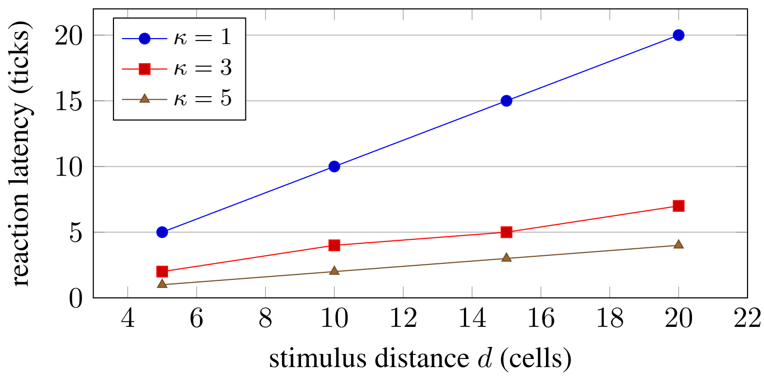

Figure 2: The internal light cone: reaction latency to a one-tick stimulus reveal. Fit over all points: latency $= 0.99\,(d/\kappa) + 0.17$, $R^2 = 0.999$ — information moves through the fields at 1.01 cells per sweep, the theoretical maximum of a radius-1 stencil.

6.6 The speed of thought

We reveal a target at distance d for one tick to a stationary agent and count think-ticks until the exact policy distribution at the agent shifts (KL > 0.1 nats from baseline). The dynamics are deterministic, so single measurements suffice. Figure 2 shows the result: latency $= 0.99\,(d/\kappa) + 0.17$ with $R^2 = 0.999$ — an internal speed of thought of 1.01 cells per sweep, exactly the stencil's light cone; at κ=1, latency literally equals distance. No trained-policy architecture produces a reaction time that is a ruler for stimulus distance; conversely, any non-local shortcut introduced into the internals would flatten this line, so the experiment is kept as a permanent locality regression test.

6.7 Non-stationarity: the pre-registered test of the thesis

A static world cannot demonstrate "online beats trained." We pre-registered three hypotheses (verbatim in the repository, written before running) on three 1000-step conditions: **regrow** (targets respawn; the episode never depletes), **drift** (targets teleport within radius 4 every 200 steps), and **swap** (at step 400, r_A and r_B exchange values). Contenders: EFI (zero training), tabular Q frozen after 2,000 episodes, the same Q continuing to learn online, and greedy-visible; the reference is a clairvoyant A^* replanning each step on a clone of the identical stochastic world. 3 seeds $\times$ 3 episodes per cell.

Table 3 summarizes. **Regrow**: EFI is the only contender in the black (+5.51), collecting 4–18$\times$ more

Table 3: Non-stationarity (1000-step episodes; lag = steps for the 20-step mean reward to re-enter 20% of its pre-shift level; "never" = no recovery within the episode).

Condition	Metric	EFI	Q frozen	Q online	Greedy
regrow	return	$\mathbf{+5.51}$	-13.02	-11.87	-5.89
	A collected / episode	**16.7**	0.9	3.9	4.1
	regret slope ($\times 10^{-2}$)	**2.2**	4.1	4.0	3.4
drift	mean lag (steps)	**46**	67	98	$7^{\dagger}$
	never recovered (of 36 shifts)	**1**	4	7	$7^{\dagger}$
	final regret	1.07	4.63	4.00	$\mathbf{0.09}^{\dagger}$
swap	adaptation lag (steps)	**5**	75	69	83
	newly-aversive pickups (pre→post)	$\mathbf{1.89 \rightarrow 0.11}$	—	—	—
	return	$\mathbf{-6.79}$	-18.01	-10.19	-9.29

targets — sustained foraging needs exploration that never goes extinct, which the entropy-regenerating epistemic term provides by construction. **Swap** is the decisive revaluation result: EFI's behavioral lag is **5** steps versus 69–83, and its pickups of the newly-aversive target drop $17\times$ — once the valence flips (within ≤ 3 contacts), a few sweeps re-propagate value under the new reward structure. *Revaluation without relearning* is exactly what separates recomputing values from having memorized them. **Drift** supports the hypothesis directionally (fewer non-recoveries and $4\times$ lower regret than either Q variant) with an adverse finding we report as pre-committed†: memoryless greedy is competitive under drift this local — radius-4 teleports keep targets near where they were, myopia is cheap, and EFI pays a small curiosity tax (return -9.07 vs. greedy's -8.09). Mild, local drift under-rewards memory; the framing survives, quantified rather than protected.

6.8 Removing the last cheat: egocentric embodiment

The world-frame controller, like most gridworld agents, silently reads its own coordinates and the world's dimensions. The egocentric variant removes both (enforced by an AST-level test: no attribute of the environment is touched): fields live on a fixed internal map, pose is dead-reckoned from an efference copy (the only feedback is *did my motor command succeed*), world boundaries are discovered by looking or bumping, and planning runs on the agent's own map with only *known* walls masked — bumping into undiscovered walls is possible and is how they are mapped. On matched seeds it is indistinguishable from the world-frame agent (return -1.00 ± 0.35 vs. -1.08 ± 0.40; success $12/12$ vs. $11/12$; first-episode protocol) — removing the GPS costs nothing. Under 10% actuation slip (moves succeed but land on a random neighbor while proprioception reports success), two local mechanisms maintain the pose: *visual odometry* — the overlap of consecutive windows aligns with the true displacement, detecting slips the tick they occur — and anchor-based template matching against the internal map for residual drift, scored only on informative cells over intersection-matched supports (both details matter; naive scoring lets uniform terrain or asymmetric visibility outvote the truth). Result: mean pose error $\mathbf{0.42}$ cells vs. 5.52 for dead reckoning alone ($13\times$; worst case 1.75 vs. 10.54 over 200-step episodes, 12 seeds).

6.9 Safety dials, verified

The exact barrier of Proposition 2 shows zero entries into a forbidden membrane ring across 500-step adversarial walks for λ spanning 0.005–0.1; action-distribution entropy decreases monotonically in pain (10k-sample estimates across the pain range); and behavior is continuous in pain — no cliff at the mode-switch threshold a discrete-flip design would carry (return difference within noise at pain 0.6 ± 10^{-3}). The failure mode that remains — an agent fully enclosed by its own barrier — is logged per tick rather than silent.

7 Limitations and honest negatives

Scope. Everything here is a discrete gridworld with noiseless, occlusion-free windows; continuous state, occlusion, and perceptual noise are untested (the belief machinery accepts soft evidence by construction, but we have not measured it). **Drift caveat.** Under mild local drift, memoryless greedy matches EFI's return (Section 6.7); the curiosity tax is real. **Pyramid.** A two-level coarse-to-fine value pyramid provably accelerates cold-start convergence, but used as an every-tick bound it *injects optimism through coarsened walls* and mildly hurt behavior ($100\% \to 90\%$ success at 50×50); it ships default-off as a cold-start accelerator only. We report this negative result because coarse-to-fine value propagation is an obvious next step for others. **Pose correction** contains drift but cannot fully undo it: evidence written at a wrong pose before the next correction stays in the map; principled repair needs confidence-weighted (deferred) mapping. **The world model does not yet plan.** Its imagination rollouts are validated but unused by control; anticipatory seeding of the value field is the natural extension. **Saturation.** ForageWorld cannot resolve $\kappa > 1$ or reward the pyramid; harder benchmarks are needed to exercise the machinery this architecture makes measurable. **No neural components** also means no representation learning: channels (walls, A, B) are given, not discovered.

8 Conclusion

Under the constraint that intelligence must be computed locally, online, and without training, the right internal objects turn out to be classical: a Bayes filter for what is out there and a linearly-solvable value recursion for what to do about it, coupled by a one-parameter risk dial and an entropy-derived curiosity reward. The reformulation keeps everything the field-based approach promised — full inspectability, per-tick bounded computation, safety by construction — while replacing its heuristics with quantities that have theorems, and its tuning folklore with dials that have units: a reach budget in cells, a thinking rate in sweeps, a speed of thought measured at 1.01 cells per sweep against a theoretical bound of 1. The agent that results is not a better-tuned reactive controller; it is a small, legible optimal-control machine that happens to be built entirely out of the same local field operations the reactive controller used — evidence that, at least at this scale, the gap between "emergent field dynamics" and "optimal control" is a change of interpretation, not of substrate.

Artifacts. Implementation (`efi/`), 213 tests including every quantitative gate above, experiment scripts, and data are in the repository; Appendix A lists per-result commands.

Ethics and reproducibility statement

The agent is a gridworld control study; no human data are involved. All headline numbers derive from seeded, deterministic runs with commands below; the non-stationarity hypotheses were written and committed before the experiment ran, and adverse findings (drift caveat, pyramid regression, curiosity tax) are reported alongside favorable ones.

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were reviewed and verified by the author.

A Reproduction commands

```
pip install -e .
pytest -q                    # 213 tests; every claim above has a gate

# Table 1 (baselines, normalized scores)
python scripts/make_baseline_table.py --episodes 40 --seeds 5 \
    --q-train-episodes 2000

# Scale invariance + Figure 1 (kappa curves) + tracking bound (Sec. 6.5)
python scripts/exp_kappa.py --episodes 20 --seeds 3 --deep-verify

# Table 2 (ablations): efi CLI, e.g.
python cli.py eval --controller field --episodes 25 --seeds 3 \
    --H 15 --W 15 [--epistemic none|frontier] [--trail 0] [--belief 0]

# Figure 2 (speed of thought)
python scripts/exp_lightcone.py

# Table 3 (non-stationarity; hypotheses in docs/EXPERIMENTS_NONSTAT.md)
python scripts/exp_nonstat.py --seeds 3 --episodes 3 --steps 1000

# Egocentric variant (Sec. 6.7)
python cli.py eval --ego 1 --episodes 10 --seeds 2
pytest tests/test_egocentric.py -q           # incl. 13x pose-error gate
```

B Hyperparameters

One configuration for all experiments. Beliefs: $\ell_{\pm}$=±12 clamped to $[-8, 8]$, prior -4, diffusion 0.14/tick, prior-relaxation 0.002/tick (both gated by static-confidence). Value: λ_0=0.02 ($\lambda \in [0.005, 0.1]$, pain gain 0.9, arousal gain 0.3), κ=3 (H+W on episode start), q_{step}=0.01, q_{trail}=0.08, q_{corner}=0.02, q_{membrane}=0.3, q_{pain}=0.3, barrier threshold 0.75. Epistemic: β_0=0.3, curiosity gain 0.5, fear gain 0.8, w_{map}=1. Valences: init $(1.0, 0.1)$, lr 0.25, clip 1.5. Schema: $3\times3\times4$ neighborhood keys, confidence threshold 0.8, hold-out every 10th transition. Egocentric: map 129^2, pose-correction snap margin $+2$, contradiction-triggered search radius 2 for 3 ticks.

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